

AP CSP | Lesson 29: Privacy & Security

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Use the stated fictional scenarios, models and simplified permission cards. Explain each decision with evidence; never enter or submit real passwords or personal details.

1. Which statement about personally identifiable information (PII) is most accurate? [1 mark]

- | | |
|---|--|
| A. Only a legal name can be PII. | B. A field must identify exactly one person by itself to be PII. |
| C. Information that describes or links to an individual can be PII, and combining fields can increase what is inferred. | D. Removing names guarantees that no personal information remains. |

Answer: _____

2. The fictional Community Shade Map only needs North/South summaries of shade ratings. Which field set best meets this purpose while limiting unnecessary collection? [1 mark]

- | | |
|---|----------------------------------|
| A. zone and shadeRating | B. name, email and exactGPS only |
| C. every available field because it might be useful later | D. password and fingerprint |

Answer: _____

3. Which sign-in method uses two different authentication factor categories? The phone step verifies possession of the registered device. [1 mark]

- | | |
|-------------------------|--|
| A. password and PIN | B. password entered twice |
| C. PIN and a second PIN | D. password and approval from the registered phone |

Answer: _____

4. A sender wants to encrypt a confidential message for Blake in the public-key model taught in class. Which pairing is correct? [1 mark]

- | | |
|---|--|
| A. Encrypt with the sender's private key; decrypt with the sender's public key. | B. Encrypt with Blake's public key; decrypt with Blake's private key. |
| C. Encrypt with Blake's private key; publish that private key for decryption. | D. Encrypt with a public key; anyone's unrelated private key can decrypt it. |

Answer: _____

5. Select exactly TWO supported statements. [1 mark]

- | | |
|---|---|
| A. A valid certificate proves that every request on the site is honest. | B. Software updates can patch known security flaws. |
| C. A familiar sender's account may have been compromised. | D. Every public wireless network is a rogue access point. |

Answer: _____

6. A malicious program records keyboard input to capture credentials. Which term most directly describes this behavior? [1 mark]

- | | | | |
|----------------|--------------------------|---------------|---------------------|
| A. aggregation | B. public-key encryption | C. keylogging | D. data compression |
|----------------|--------------------------|---------------|---------------------|

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

The fictional map needs only North/South aggregate shade priorities. Proposed fields are zone, shadeRating, name, email and exactGPS. Choose the two necessary fields, explain their roles, identify unnecessary collection, and explain one privacy risk that can remain after publishing only zone summaries.

8. [4 marks]

Compare password + PIN with password + verified phone approval. Classify the password and PIN, classify the phone step, identify which combination qualifies as MFA, and explain one limitation of MFA. No real credentials are needed.

9. [4 marks]

For a confidential message sent to Blake, identify the public-key encryption key and decryption key, explain which key Blake may share and which stays secret, and contrast this with symmetric encryption.

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Written responses | Questions 10-12

10. [4 marks]

A fictional message from a familiar account says, “Open this form now and send your password to keep using the map.” A separate untrusted download claims to be a required map update. Give two distinct risks and a suitable response to each.

11. [4 marks]

A student says, “This connection is encrypted and has a valid digital certificate, so the request to reveal my password must be safe.” Explain the role of encryption, the role of a certificate authority, the flaw in the claim, and a safer response.

12. [4 marks]

Explain what a rogue access point is, why “all public Wi-Fi is rogue” is incorrect, how reviewing app permissions can protect privacy, and why software updates matter.
